

**Central Valley Regional Water Quality Control Board**

25 March 2013

Steve Bonaker  
Department of the Navy  
Naval Air Station  
700 Avenger Avenue  
Lemoore, CA 93246

**REVISED MONITORING REPORTING PROGRAM UNDER ORDER R5 2002-0062,  
LEMOORE NAS WWTF, KINGS COUNTY (WDID 5D160700001, RM 131463)**

Your 17 February 2012 letter requests termination of wildlife monitoring requirements set forth in Waste Discharge Requirements Order R5-2002-0062 (WDRs). You submitted evidence of pond water selenium concentrations below 1-3 parts per billion (ppb) (USFWS background level) and avian egg selenium concentration of 1.26 part per million.

We consulted with Dr. Andrew Gordus of the California Department of Fish and Wildlife, and have concluded that reduced monitoring, utilizing a tiered approach, is warranted for wildlife monitoring. If the pond water selenium concentrations (all ponds averaged) remain below a quarterly average of 2 ppb, then only dead bird monitoring (per WDRs) should occur to avert any botulism outbreak. If the quarterly average exceeds 2 ppb, then nest surveys should be conducted during the breeding season (March to August) in accordance with the revised monitoring and reporting program discussed below. If nests are present, then egg collection and analyses for selenium concentration should occur. Based upon egg selenium concentrations, additional mitigation measures may be needed.

We circulated a draft revised Monitoring and Reporting Program (MRP) for your review and comment on 9 March 2012 to which you have no comments.

A revised MRP is attached, modified as outlined above, and signed by the Executive Officer, which is now a part of your Waste Discharge Requirements. If you have any questions on this matter please contact Anthony Toto at 559-445-6278 or [atoto@waterboards.ca.gov](mailto:atoto@waterboards.ca.gov).



WARREN W. GROSS  
Senior Engineering Geologist

Enclosure: Revised MRP

cc: Dr. Andrew Gordus, Department of Fish and Wildlife, Fresno

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD  
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. R5-2002-0062  
REVISED 22 MARCH 2013

FOR  
U.S. DEPARTMENT OF THE NAVY  
LEMOORE NAVAL AIR STATION  
WASTEWATER TREATMENT FACILITY  
KINGS COUNTY

This Monitoring and Reporting Program (MRP) is required pursuant to California Water Code section 13267. The Discharger shall not implement any changes to this MRP unless a revised MRP is adopted by the Regional Board or issued by the Executive Officer. All samples should be representative of the volume and nature of the discharge or matrix of material sampled. The time, date, and location of each sample shall be recorded on the sample chain of custody form. Attachments B and C indicate the sampling locations. All analyses shall be performed in accordance with Standard Provisions, Provisions for Monitoring.

**INDUSTRIAL WASTEWATER TREATMENT FACILITY  
EFFLUENT MONITORING**

The Discharger shall collect effluent samples from the industrial wastewater treatment facility at a point following the final treatment unit, but before it combines with domestic sewage. Effluent monitoring shall include the following:

| <u>Constituent</u>           | <u>Units</u> | <u>Type of Sample</u> | <u>Sampling Frequency</u>    |
|------------------------------|--------------|-----------------------|------------------------------|
| Flow                         | mgd          | Metered               | Continuous                   |
| Average Daily Flow           | mgd          | Calculated            | Daily                        |
| pH                           | pH units     | Grab                  | Weekly                       |
| EC                           | umhos/cm     | Grab                  | Weekly                       |
| Settleable Solids            | mL/L         | Grab                  | Weekly                       |
| Total Suspended Solids       | mg/L         | Grab                  | Weekly                       |
| Oil and Grease               | mg/L         | Grab                  | Twice/Monthly <sup>1</sup>   |
| Dichloromethane              | µg/L         | Grab                  | Twice/Monthly <sup>1,2</sup> |
| Copper                       | mg/L         | 8-hr composite        | Monthly                      |
| Nickel                       | mg/L         | 8-hr composite        | Monthly                      |
| Total Chromium               | mg/L         | 8-hr composite        | Monthly                      |
| Zinc                         | mg/L         | 8-hr composite        | Monthly                      |
| Silver                       | mg/L         | 8-hr composite        | Monthly                      |
| Lead                         | mg/L         | 8-hr composite        | Monthly                      |
| Cadmium                      | mg/L         | 8-hr composite        | Monthly                      |
| Total Cyanide                | mg/L         | Grab                  | Monthly                      |
| Trichloroethane <sup>3</sup> | µg/L         | Grab                  | Monthly                      |
| Priority Pollutants          | µg/L         | Grab                  | Annually <sup>4</sup>        |

<sup>1</sup> Nonconsecutive weeks

<sup>2</sup> After one year, frequency may be reduced to monthly

<sup>3</sup> 1,1,1-Trichloroethane and 1,1,2-Trichloroethane

<sup>4</sup> October

## DOMESTIC WASTEWATER TREATMENT FACILITY INFLUENT MONITORING

The Discharger shall collect influent samples at the headworks prior to any treatment of waste. Influent monitoring shall include at least the following:

| <u>Constituent</u>            | <u>Units</u> | <u>Type of Sample</u>       | <u>Sampling Frequency</u> |
|-------------------------------|--------------|-----------------------------|---------------------------|
| Maximum Daily Flow            | mgd          | Continuous                  | Continuous                |
| Average Daily Flow            | mgd          | Computed                    | Daily <sup>1</sup>        |
| Monthly Average Flow          | mgd          | Computed                    | Monthly                   |
| Settleable Solids             | mL/L         | Grab                        | Weekly                    |
| pH                            | pH units     | Grab                        | Weekly                    |
| BOD <sub>5</sub> <sup>2</sup> | mg/L         | 8-hr Composite <sup>3</sup> | Monthly <sup>4</sup>      |
| TSS <sup>5</sup>              | mg/L         | 8-hr Composite <sup>3</sup> | Monthly <sup>4</sup>      |

<sup>1</sup> Sample frequencies referenced hereafter in this program as daily shall not include weekends or holidays.

<sup>2</sup> Five-day, 20°C biochemical oxygen demand

<sup>3</sup> A representative composite sample may be obtained by collecting four samples at uniform intervals over an eight-hour period.

<sup>4</sup> Quarterly after one year (January, April, July, and October)

<sup>5</sup> Total Suspended Solids

## EFFLUENT MONITORING

The Discharger shall collect effluent samples at a point in the system following treatment and before discharge to evaporation Pond 1. Effluent samples shall be representative of the volume and nature of the discharge. Effluent monitoring shall include the following:

| <u>Constituent</u>                        | <u>Units</u> | <u>Type of Sample</u> | <u>Sampling Frequency</u>    |
|-------------------------------------------|--------------|-----------------------|------------------------------|
| Settleable Solids                         | mL/L         | Grab                  | Weekly                       |
| pH                                        | pH Units     | Grab                  | Weekly                       |
| EC <sup>1</sup>                           | µmhos/cm     | Grab                  | Monthly                      |
| BOD <sub>5</sub>                          | mg/L         | 8-hr Composite        | Monthly <sup>2</sup>         |
| TSS                                       | mg/L         | 8-hr Composite        | Monthly <sup>2</sup>         |
| Total Dissolved Solids (TDS) <sup>3</sup> | mg/L         | 8-hr Composite        | Monthly <sup>2,4</sup>       |
| Nitrate (as N)                            | mg/L         | 8-hr Composite        | Semiannually <sup>5</sup>    |
| Total Kjeldahl Nitrogen (TKN)             | mg/L         | 8-hr Composite        | Semiannually <sup>5</sup>    |
| Total Nitrogen                            | mg/L         | Calculated            | Semiannually <sup>5</sup>    |
| Antimony                                  | µg/L         | Grab                  | Annually <sup>6</sup>        |
| Molybdenum                                | µg/L         | Grab                  | Annually <sup>6</sup>        |
| Arsenic                                   | µg/L         | Grab                  | Annually <sup>6</sup>        |
| Selenium                                  | µg/L         | Grab                  | Twice/Monthly <sup>2,4</sup> |
| General Minerals <sup>7</sup>             | mg/L         | Grab                  | Monthly <sup>2,4</sup>       |

- 1 Conductivity at 25°C
- 2 After one year, frequency may be reduced to quarterly (i.e., January, April, July, and October).
- 3 TDS referenced hereafter in this program shall be determined using EPA Method No. 160.1 for combined organic and inorganic TDS and EPA Method No. 160.4 for inorganic TDS or equivalent analytical procedures specified in 40 CFR Part 136.
- 4 Shall coincide with EC sampling
- 5 April and October
- 6 October
- 7 General Minerals as referred to in this program shall include the constituents in the General Minerals Analyte List presented below. Using proper sampling methods and appropriate sample containers is critical in obtaining valid results for general minerals analyses. Follow laboratory directions and secure sample containers as appropriate for requesting analyses for general minerals.

### General Minerals Analyte List

|                                     |                                   |           |
|-------------------------------------|-----------------------------------|-----------|
| Alkalinity (as CaCO <sub>3</sub> )  | Carbonate (as CaCO <sub>3</sub> ) | Manganese |
| Aluminum                            | Chloride                          | Phosphate |
| Bicarbonate (as CaCO <sub>3</sub> ) | Hardness (as CaCO <sub>3</sub> )  | Potassium |
| Boron                               | Iron                              | Sodium    |
| Calcium                             | Magnesium                         | Sulfate   |

### POND MONITORING

The Discharger shall monitor each of its stabilization and evaporation ponds for the parameters specified below. Freeboard shall be monitored on all ponds in use to the nearest one tenth of a foot. Pond monitoring shall include at least the following:

| <u>Constituent/Parameter</u> | <u>Units</u> | <u>Type of Sample</u> | <u>Sampling Frequency<sup>1</sup></u> |
|------------------------------|--------------|-----------------------|---------------------------------------|
| Selenium                     | mg/L         | Grab                  | Twice/Monthly                         |
| Dissolved Oxygen (DO)        | mg/L         | Grab <sup>2</sup>     | Weekly                                |
| Pond Depth                   | feet         | Observation           | Weekly                                |
| Freeboard                    | feet         | Observation           | Weekly                                |

- 1 If results of monitoring appear to violate discharge specifications, but monitoring frequency is not sufficient to validate violation or indicate a violation and potential upset of the treatment process (e.g., less than minimum dissolved oxygen concentration), the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.
- 2 Samples shall be collected at a depth of one foot from each pond in use, opposite the inlet, and analyzed for DO. Samples shall be collected between 0700 and 0900 hours. Time of sampling shall be reported. If DO results for any pond in use indicate noncompliance with the discharge specification, the Discharger shall implement corrective measures as specified in the O&M manual and monitor said pond daily until its DO stabilizes above 1 mg/L.

In addition, the Discharger shall inspect the condition of the ponds once per week and write visual observations in a bound logbook. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the ponds (e.g., dark sparkling green, dull green, yellow, gray, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. If the Discharger finds itself in violation of Domestic WWTF Discharge Specifications C.3, C.4, C.5, C.6, C.9, or C.10 the Discharger shall briefly explain the action taken or to be taken to correct the violation. The Discharger shall certify in each November monitoring report that it is in compliance with Domestic WWTF Discharge Specification C.13.

## SLUDGE MONITORING

**Within 270 days of the adoption of this Order**, the Discharger shall characterize sludge quality from its domestic treatment ponds for the following constituents:

|            |          |          |           |
|------------|----------|----------|-----------|
| Arsenic    | Lead     | Zinc     | Potassium |
| Cadmium    | Mercury  | Sulfate  | Sodium    |
| Molybdenum | Nickel   | Chloride |           |
| Copper     | Selenium | Boron    |           |

To characterize sludge, the Discharger shall collect a composite sample from each of its ponds. For guidance in sampling, the Discharger should consult EPA's *POTW Sludge Sampling and Analysis Guidance Document*, August 1989. The Discharger shall submit the results of the sludge characterization to the Regional Board by the **1<sup>st</sup> day of the second month** following sampling.

## REVISED WILDLIFE MONITORING

The Discharger shall inspect each domestic WWTF pond weekly for dead birds. Inspections shall be increased to daily at any pond when the water depth is less than two (2) feet and for all ponds when a botulism or fowl cholera outbreak is occurring in the area, as confirmed by the California Department of Fish & Wildlife (CDFW). Once DFW confirms the said outbreak is over, the Discharger may reduce the frequency of pond inspections back to weekly.

If the pond monitoring water selenium concentrations remain below a quarterly average of 2 ppb, then only dead bird monitoring should occur to avert a botulism outbreak.

If the quarterly average concentrations of selenium in pond water (all ponds averaged together) exceeds 2 ppb, then wildlife monitoring at the domestic WWTF ponds shall be conducted by or under the direct supervision of a qualified wildlife biologist with a permit (or able to obtain a permit) from the U. S. Fish and Wildlife Service to collect eggs according to the following:

- 1) Bird counts shall be conducted twice monthly.
- 2) If birds nest at the ponds, then breeding bird surveys shall be conducted twice monthly until completion (typically from March through August and must include counts of nests, eggs and young. Nests shall be flagged, and five (5) random recurvirostrid eggs from five separate nests shall be collected and shall be chemically analyzed for total recoverable selenium and moisture content.

## REPORTING

The Discharger shall report monitoring data and information as required in this MRP and as required in the Standard Provisions and Reporting Requirements. All reports submitted in response to this MRP shall comply with the signatory requirements in Standard Provisions, General Reporting Requirements B.3. Daily, twice weekly, weekly, twice monthly, and monthly monitoring data shall be reported in monthly monitoring reports. Monthly monitoring reports shall be submitted to the Regional Board by the **1<sup>st</sup> day of second month following sampling**. Quarterly monitoring reports shall be submitted by **1<sup>st</sup> day of second month after the calendar quarter**.

In each monitoring report, the Discharger shall discuss the compliance record for the reporting period. If violations have occurred during the reporting period, the report shall also discuss the corrective actions taken and/or planned to bring the discharge into full compliance with this Order. Monitoring data and/or discussions submitted concerning WWTF performance must also be signed and certified by the chief plant operator. When reports contain laboratory analyses performed by the Discharger and the chief plant operator is not in the direct line of supervision of the laboratory, reports must also be signed and certified by the chief of the laboratory.

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements. If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

All wildlife monitoring (if required) shall be submitted to the Regional Board as follows:

| <u>Reporting Period</u>                  | <u>Draft</u>             | <u>Final</u> |
|------------------------------------------|--------------------------|--------------|
| October-March                            |                          | 1 May        |
| <u>April-September</u>                   | 20 December <sup>1</sup> | 20 February  |
| <sup>1</sup> Five copies for peer review |                          |              |

The Discharger may also be requested to submit an annual report to the Regional Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss any corrective actions the Discharger takes or plans to take to bring the discharge into full compliance with the waste discharge requirements.

**By 1 February of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

1. The names, certificate grades, and general responsibilities of all persons in charge of wastewater treatment and disposal.
2. The names and telephone numbers of persons to contact regarding the WWTF for emergency and routine situations.
3. A statement certifying when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration (Standard Provision C.4). Reports submitted concerning WWTF performance must also be signed and certified by the chief plant operator. When reports contain laboratory analyses performed by the Discharger and the chief plant operator is not in the direct line of supervision of the laboratory, reports must also be signed and certified by the chief of the laboratory.
4. A statement whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment facility as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
5. The results of an annual evaluation conducted pursuant to Standard Provision E.4 and a figure depicting monthly average discharge flow for the past five years.

6. The most recent LNAS annual water supply report.
7. A summary of sludge monitoring, including
  - a. Annual sludge production in dry tons and percent solids.
  - b. A schematic diagram showing sludge handling facilities and solids flow diagram.
  - c. A description of disposal methods, including the following information related to the disposal methods used at each WWTF. If more than one method is used, include the percentage of annual sludge production disposed of by each method.
    - i. For **landfill disposal**, include: (a) the Order numbers of WDRs that regulate the landfill(s) used, (b) the present classifications of the landfill(s) used, and (c) the names and locations of the facilities receiving sludge.
    - ii. For **land application**, include: (a) the locations of the site(s), and (b) the Order numbers of any WDRs that regulate the site(s).
    - iii. For **incineration**, include: (a) the names and location of the site(s) where sludge incineration occurs, (b) the Order numbers of WDRs that regulate the site(s), (c) the disposal method of ash, and (d) the names and locations of facilities receiving ash (if applicable).
    - iv. For **composting**, include: (a) the location of the site(s), and (b) the Order numbers of any WDRs that regulate the site(s).
8. A summary and discussion of the measures taken to eliminate dichloromethane from the industrial WWTF discharge.
9. A summary and discussion of the compliance record for the reporting period. If violations have occurred, the report shall also discuss the corrective actions taken and planned to bring the discharge into full compliance with this Order.

The Discharger shall implement the above monitoring program on the first day of the month following adoption of this Order.

Ordered by: Clay L. Rodgers  
for Pamela C. Creedon, Executive Officer

3/25/13

(Date)